AEP as the Agentic-Container OS

High-Level Design (HLD) · v1.1 · Status: Proposed · Branch: `docs/dual-lane-governance-proposal` · Companion: `aep-lld-v1.1.html`
Supersedes: v1 (dual-lane governance proposal, `docs/draft-proposal.html` v1)

1. Purpose and scope

This document consolidates the design decisions reached in the working discussion into one reference: how the Autonomous Engineering Platform (AEP) is built and deployed portably, and how it governs two kinds of autonomous execution — fast local **ralph-loop** lanes and sandboxed **ESF** (Engineering Software Factory) pipelines — on a shared set of core services.

It is a design, not a record of finished work. Section 9 states exactly what exists in the repository today and what is still to be built; reading the rest of the document as "already implemented" would be wrong.

2. Decision log

#	Decision	Rationale	Status
D1	Orchestration tier is a single Go binary, `aep`; NeuroStrata stays in Rust.	Static cross-compiled binaries need no Python on Windows/embedded hosts; `golang.org/x/sys` gives Job Objects, `flock` / `LockFileEx` and process groups. Python's one real edge (`ast`-based test classification in `mechanical_verify.py`) is narrow.	BUILT (CORE)
D2	Artifact targets: `dist/bin/aep-{linux,darwin}-{amd64,arm64}`, `dist/vm/linux-{amd64,arm64}`, plus OCI image/context.	Covers Linux servers, the 8 GB Muse ARM64 VM, Apple Silicon and Intel Macs, and any Docker/Podman host. Native Windows is explicitly out of the target list (WSL2 or container only).	BUILT
D3	Two execution lanes: Tier 1 ralph-loop, Tier 2 ESF.	Local worktree lanes are fast but share the host kernel; ESF adds microVM isolation and durable workflows for untrusted or high-blast-radius work.	PLANNED
D4	AEP is the "container OS": five shared core services on the substrate; lanes are ephemeral containers above it.	One memory, one ledger, one rate limiter, one policy for every lane — no per-lane drift.	PLANNED
D5	Runtime: one Colima VM (`vz` + `virtiofs`) sized 12 CPU / 48 GB RAM / 220 GB disk, multiplexing containers. Not one VM per lane.	Dynamic sharing of CPU/RAM across lanes; per-profile VMs only where kernel-level isolation is required.	PLANNED
D6	`beads` is the single macro task ledger. The Superpowers SDD ledger (`.superpowers/sdd/<plan>/progress.md`) stays a per-plan micro ledger and does not replace it.	Different lifetimes and purposes: durable cross-session backlog vs. ephemeral compaction-recovery record.	PLANNED
D7	ESF workflow IDs are keyed to beads issue IDs (`esf-beads-<ID>`).	Prevents orphan factory runs and dual-backlog drift.	PLANNED
D8	Factory-tagged tasks degrade explicitly when Temporal/CubeSandbox are unreachable (quarantine, or fenced local lane with escalated review) — never hang.	Removes the load-bearing failure found in review: the loop stalling on infrastructure that does not exist on most hosts.	PLANNED

3. Goals and non-goals

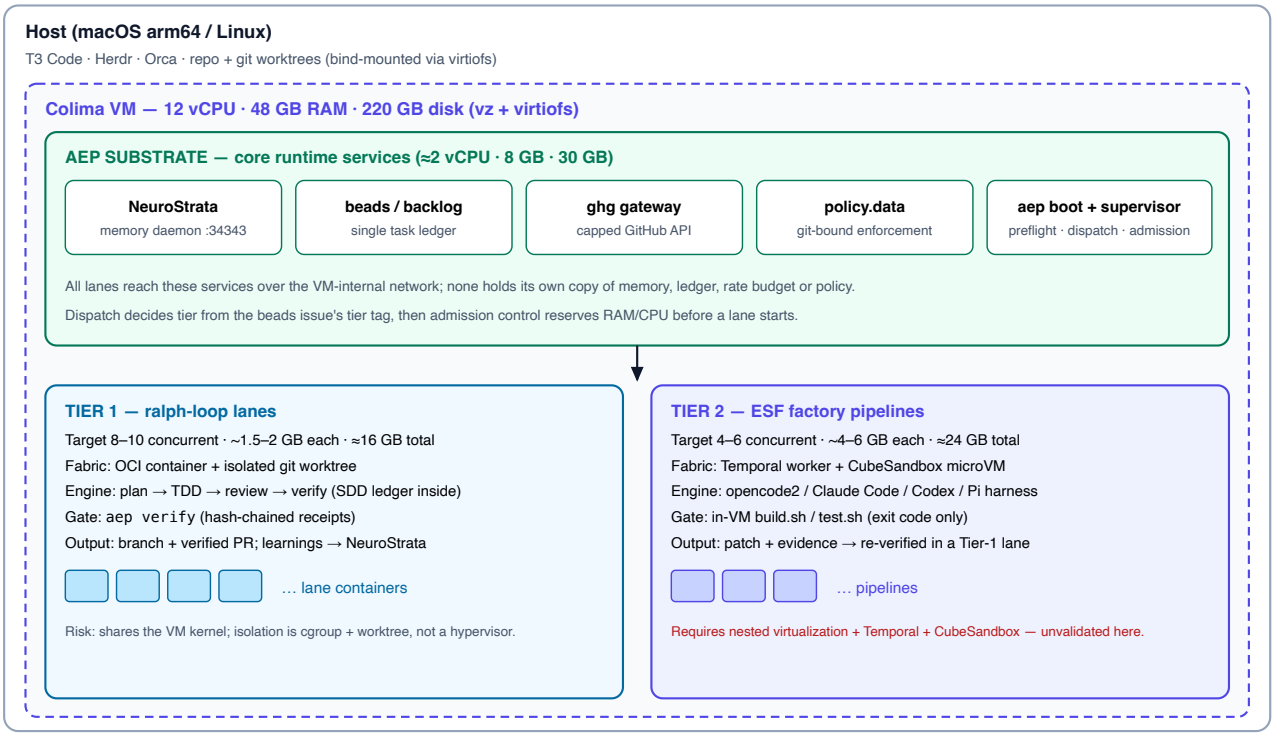
Goals

- One-command install and run on Linux (x86_64 and ARM64), the Muse 8 GB VM, and macOS (Intel and Apple Silicon).
- Many parallel lanes under one governance plane with bounded, predictable resource use.
- Deterministic verification decides success; an agent saying "done" never does.
- Fail closed: no autonomous run starts if the environment is unfit.

Non-goals (v1.1)

- Native Windows binaries or service wrappers.
- macOS guest VM images (macOS runs the native binary under `launchd`).
- Replacing `beads` or NeuroStrata.
- Modifying the vendored `reference/esf` tree (it changes only by re-vendoring).

4. System architecture



5. Components

5.1 Core runtime services (substrate)

Service	Responsibility	Key constraint
NeuroStrata	Hybrid vector + full-text memory (`ladybug.db`, nomic-embed-text-v1.5). Retrieval before planning, inscription after verification.	Embedding model cache pre-seeded; `HF_HUB_OFFLINE=1`; DB backed up by change-detected watchdog.
beads / backlog	Single canonical task ledger; carries the tier tag and the foreign key used by ESF workflow IDs.	Telemetry off (`bd metrics off`); Dolt remote host allowlisted if egress is blocked.
ghg	Counted, coalescing GitHub gateway: 3,500/5,000 core req/h self-cap, one poller per watched run.	Governs only callers that use it; direct `gh` remains an ungated vector.
policy.data	Env-variable scrub fragments, sensitive-path patterns, hardware profile; bound to the committed git blob.	Edits land via PR; boot refuses a policy that differs from `HEAD`.
aep boot / supervisor	Fail-closed preflight, dispatch, capacity admission, periodic ticks.	Exit non-zero rather than start in a degraded state.

5.2 Execution lanes

	Tier 1 · ralph-loop	Tier 2 · ESF
Intended work	Features, fixes, refactors, tests, docs	Untrusted dependencies, CVE remediation, multi-repo or high-blast-radius change
Isolation	Container + git worktree + process group / Job Object	Hardware-isolated microVM per run, destroyed on every exit path
Orchestration	Scheduler tick (default 2 h), ≤6 iterations, 3-strike stop	Temporal durable workflow with replay-safe activities
Verification	`aep verify`	Repo `build.sh` / `test.sh` in the microVM (target: bake `aep verify` into the template)
Human gate	PR review; circuit breaker	`factory review --approve/--reject`

6. Resource model (12 CPU / 48 GB / 220 GB)

Resource	Total	Substrate	Tier 1	Tier 2
vCPU	12	2 reserved	4–6 shared	4–6 priority
RAM	48 GB	8 GB	16 GB	24 GB
Disk	220 GB	30 GB (models, DB)	70 GB (worktrees, caches)	120 GB (microVM images, artifacts)

> **Capacity is memory-bound, not count-bound.**
> Six Tier-2 pipelines at 4 GB fit the 24 GB budget; six at 6 GB do not (36 GB). The "4–6 pipelines" and "8–10 lanes" figures are therefore *ceilings under typical sizing*, not guarantees. The design enforces this with admission control that reserves memory per lane before start (LLD §6), rather than a fixed lane count. Per-lane footprints are estimates from capacity reasoning; **they have not been benchmarked on this machine**, and no container runtime is currently installed on the host to measure with.

Reference sizing for other targets: the Muse 8 GB VM runs the substrate plus at most 2 lanes (`AEP_MAX_LANES=2`, 384 MiB floor, ONNX threads 2); Tier 2 is not expected on Muse.

7. Platform and artifact coverage

Artifact	Covers	Does not cover
`dist/bin/aep-linux-amd64`	Linux x86_64 servers, cloud VMs, CI runners	32-bit x86; native Windows
`dist/bin/aep-linux-arm64`	Muse ARM64 VM, Graviton/Ampere, Raspberry Pi 4/5 (64-bit)	32-bit ARM
`dist/bin/aep-darwin-arm64`	Apple Silicon Macs	Intel Macs (Rosetta only)
`dist/bin/aep-darwin-amd64`	Intel Macs	—
`dist/vm/linux-{amd64,arm64}`	Cloud-init + systemd bundles with per-arch hardware profile	Non-systemd distros
OCI image / `dist/oci` context	Any Docker/Podman host; Kubernetes-style schedulers	Hosts with no container runtime

8. Dispatch and lifecycle

1. **Intake.** Work is a beads issue carrying a tier tag (`local` or `factory`) and acceptance criteria.
2. **Preflight.** `aep boot` must pass: binaries, NeuroStrata answering, standing guards present, policy blob matches `HEAD`, memory floor met.
3. **Admission.** Dispatcher reserves RAM/CPU for the lane against the profile budget; if it cannot, the task waits (bounded) rather than oversubscribing.
4. **Execute.** Tier 1: claim worktree, start lane container, run plan/TDD/review/verify. Tier 2: submit `factory run` with workflow ID `esf-beads-<ID>`.
5. **Verify.** Both lanes emit SHA-256 receipts. ESF patches are applied in a clean Tier-1 lane and re-verified before any PR.
6. **Close.** PR merges, `bd close`, learnings inscribed to NeuroStrata, lane and reservation released.

9. Governance invariants

- No lane starts unless `aep boot` passes (fail closed).
- All GitHub traffic from lanes goes through `ghg`.
- `policy.data` must equal the committed blob; lanes run with credential-named env vars scrubbed.
- ESF refuses to run with `hardening.acknowledge_open_egress = true` in a governed deployment (the upstream default for a fresh `factory init` is `true`, so AEP must assert the opposite).
- Model tiering: ralph lanes at `standard` / `measured`; `frontier` / `thorough` reserved for factory runs.
- A task has exactly one ledger identity (the beads ID) across both lanes.

10. Implementation status (as of this document)

BUILT in `develop`

- `cmd/aep` with `version`, `info`, `boot`, `verify`, `supervisor`, `lock-test`.
- `internal/platform/{memory,lock,process}` for Linux, macOS and Windows; `internal/verifier/runner` (allowlist discovery, tail-capped output, SHA-256 receipts, memory preflight, budgets).
- `Makefile` targets `binaries`, `vm`, `oci`; `scripts/package-vm.sh`, `scripts/package-oci.sh`; `Containerfile`, `docker-compose.yml`.
- Root CI runs `gofmt`, `go vet`, `go test`.

NOT YET BUILT

- `aep boot` today checks only the memory floor and platform; the NeuroStrata/guard/gh-auth/policy checks still live in `scripts/boot-gate.sh`.
- `aep supervisor` today runs boot gate + one verification pass per tick; it does not dispatch lanes or ESF runs.
- Lane coordination remains in `scripts/lane-coord.sh` (Linux `flock`); no Go lane manager, no admission control, no beads adapter, no dispatcher, no ESF client, no Colima profile or compose topology for the substrate.
- The "unified verifier inside the CubeSandbox template" and NeuroStrata hooks in the ESF harness are design intent only.

11. Risks and open questions

Risk / question	Why it matters	Cheapest check
HIGH Tier 2 needs KVM inside the Colima VM	CubeSandbox microVMs require nested virtualization. On Apple Silicon this depends on chip generation, macOS version and Colima support. If unavailable, Tier 2 cannot run on the Mac at all.	Start Colima with nested virtualization enabled and run `ls /dev/kvm` in the VM.
HIGH CubeSandbox and Temporal are external	ESF discovers an existing CubeSandbox; it never installs one. Neither exists on this host.	`factory doctor` against the target environment.
MED Lane footprints are unmeasured	Admission thresholds depend on real RSS of ralph and ESF lanes.	Install Colima, run 3 representative lanes, record peak RSS.
MED Tier-1 isolation is weaker than it sounds	Lane containers share one VM kernel; a hostile task in Tier 1 is not contained like Tier 2.	Keep untrusted work routed to Tier 2; make the tier tag mandatory.
MED Boot parity gap	Moving to `aep boot` without the daemon/guard/policy checks would silently weaken governance versus `boot-gate.sh`.	Port checks with tests before retiring the shell gate.
LOW Singular/plural docs path	This document lives in `docs/plan/`; the repo already has `docs/plans/` and `docs/superpowers/plans/`.	Decide on one location.

12. Review status

The v1 → v1.1 review applied the paperthin lenses (`hate`, `ssotize`, `modelchk`, `factchk`, `debloat`) as a single-author pass. It surfaced the dispatch-stall root objection, the dual-ledger drift risk and the egress-default fact. It was **not an independent multi-reviewer panel**, and its "full approval" wording should be read as "no remaining objections from that pass". An independent adversarial run should precede production activation, and it should specifically attack the nested-virtualization assumption above.

AEP HLD v1.1 · generated from the design discussion on 2026-11-08 · see the LLD for interfaces, data formats, sequencing and the task breakdown.